PROJECT OVERVIEW

Power Glove Vision Project Overview

Architecture, controls, security, deployment, and project status.

2026 EDITION / IAIN BENNETT / MIT LICENSE

PowerGlove Vision lets you play RetroPie games by moving your hand in front of a camera connected to an Arduino UNO Q. Use your bare hand or a plain glove; there are no glove electronics to build. The UNO Q tracks your movements and sends controller input to a Raspberry Pi, where RetroArch sees a virtual gamepad named **PowerGlove Vision**.

The project includes eleven profiles: nine reusable Programs A-I and dedicated controls for Bad Street Brawler and Super Glove Ball. RetroPie can select a profile automatically when you launch a registered game. Glove Academy mode lets you practise without sending input to the cabinet.

The cabinet currently uses `lr-fceumm` as its default NES emulator. `lr-nestopia` has not been tested with PowerGlove Vision. Native Power Glove support remains planned work. The software preserves analogue and finger measurements for that future integration, but the current gamepad path does not provide every original glove feature.

Choose a guide

User manuals

You want to...	Read...
Install both devices and play your first game	Installation Guide
Find a command or connection reminder	Quick Reference
Learn a game's gestures and try a short challenge	Game and gesture guide
Choose or experiment with Programs A-I	Programs A-I manual
Recognize matrix animations and letters	Matrix display guide

Technical documentation

You want to...	Read...
Understand components and data flows	Architecture
Change settings or look up command flags	Configuration Reference
Understand network and pairing boundaries	Security policy
Change the project or its documentation	Contributing guide
Check dependency provenance or release history	Third-party components and Changelog

Quick start

Prepare your UNO Q with Arduino App Lab and use an existing RetroPie installation. Connect both to the same trusted network and attach the camera through a powered USB hub. Keep a physical controller available for RetroArch setup.

The commands in the Installation Guide select the latest published stable release. Use the same release on both devices. Download `install-uno-q.sh` and `install-retropie.sh` from that published release. Run the first on the UNO Q and the second on RetroPie as your normal login user. Each verifies its package and requests sudo access when needed. The UNO installer includes the Arduino sketch, early-start helper, and shutdown helper; no separate App Lab import or helper installation is needed.

Follow the Installation Guide for copyable commands, pairing, calibration, and your first game. Both scripts also support `--check` and repeatable updates while preserving personal settings. Installer assets must be published before the release download commands become available.

Controls

Calibration records the resting hand position that the app treats as the centre of movement. Move away from that position to give a direction and return to it to release that direction. Recalibrate after moving the camera or changing your playing position. Some profiles replace ordinary hand movement with wrist steering or other controls, as shown below.

Across the profiles, hold a **V sign** for about 0.7 seconds to send Start and a **thumbs-up with the other fingers closed** to send Select. These poses suppress A/B attacks; some profiles can still generate directional or auxiliary input, so keep your hand near its resting position while using them.

Programs A-I

These reusable mappings produce ordinary NES controls. You do not need to launch Bad Street Brawler first. The table describes controller output; its effect depends on the game. A pulsed button repeatedly presses and releases.

Program	Movement	Actions and special gestures
A - Pinball	Ordinary movement is disabled.	Index curl sends A; thumb curl sends Up; wrist roll sends B. Pulling back toggles combined flippers.
B - Joust	Move your hand left or right.	Index or middle curl pulses A; thumb curl holds B.
C - Gyruss	Roll your wrist left or right.	A straight index finger holds A; pulling back sends B. Use the game's Attack Control B mode.
D - Challenge	All four hand-movement directions are reversed.	Thumb curl sends A; index curl sends B.
E - Defender II	Move your hand in four directions.	Thumb curl sends A; wrist roll sends B; ring-finger curl rapidly alternates left and right.
F - Sesame Street	Ordinary directional output is disabled.	Moving an open hand away from centre sends A; closing all fingers sends B.

Program	Movement	Actions and special gestures
G - Gun Smoke	Move your hand in four directions; wrist roll adds left or right.	Index curl sends A; pushing forward sends B. Combine them for A+B. Thumb and ring-finger curl together suppress D-pad and A/B output.
H - General	Move your hand in four directions.	Index curl pulses A; thumb curl pulses B.
I - Knight Rider	Roll your wrist to steer; lower your hand to brake.	Index curl sends Up for acceleration; pushing sends Up+A for turbo; thumb curl sends B.

Programs A, D, and H have no default ROM assignment. Choose one on Dashboard to try it, then register the exact game filename if you want automatic selection. The Programs manual includes illustrations; the Game and gesture guide adds objectives and tips.

Bad Street Brawler

Gesture	Controller output
Move your hand left, right, up, or down	Corresponding D-pad direction
Curl your thumb	Pulsed B
Curl your middle finger	A+B
Roll your wrist left or right	A plus that direction
Push toward the camera	Glove Zap: short simultaneous Left + Right pulse

Push toward the camera for Glove Zap, then return to your starting distance before trying again. Bad Street Brawler needs its game-specific emulator setting; see the configuration reference.

Super Glove Ball

Gesture	Controller output
Move your hand left, right, up, or down	Corresponding D-pad direction
Curl your index finger	A
Curl your thumb	B

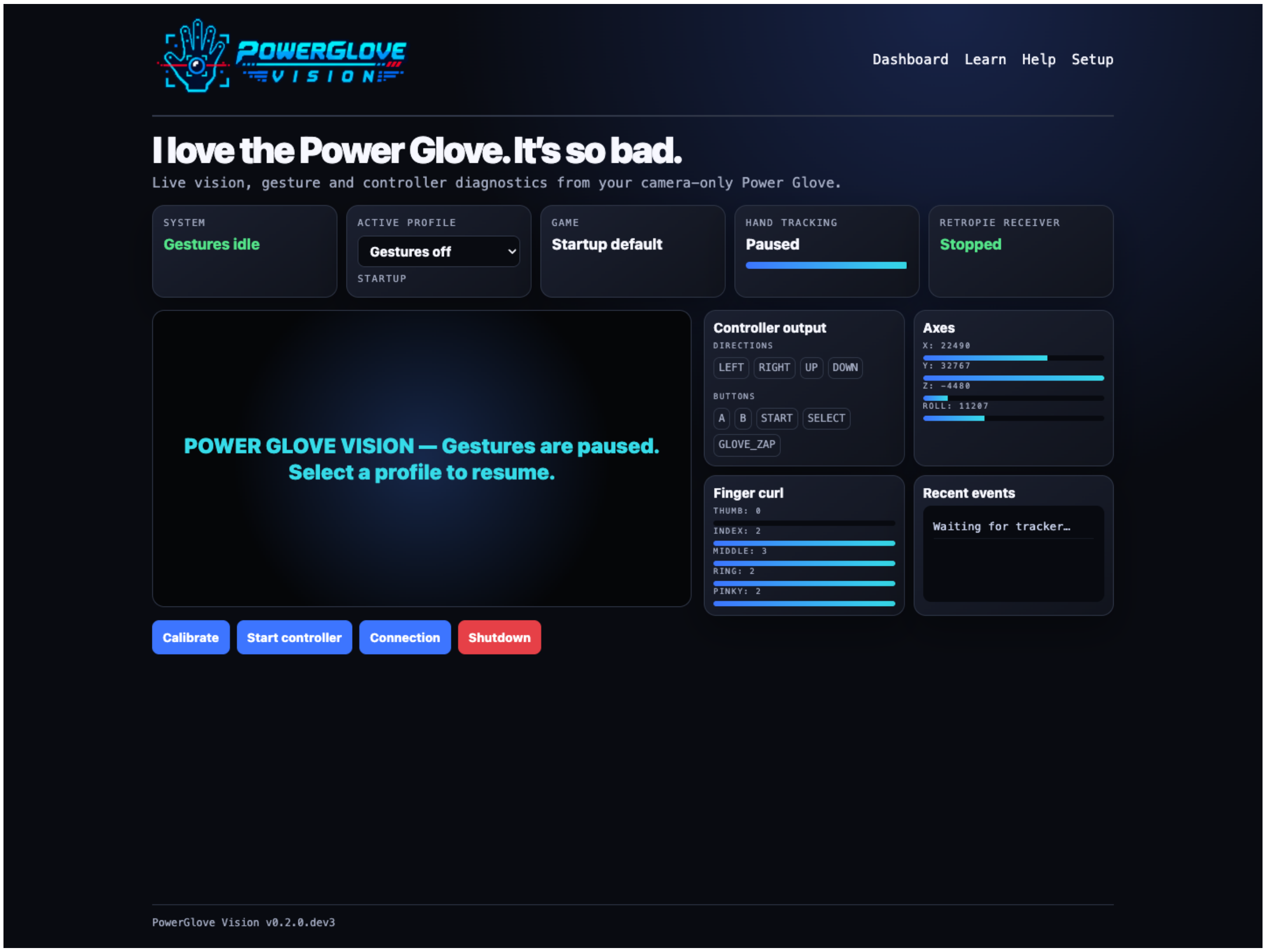
The protocol also carries hand position, estimated depth, wrist roll, and finger measurements for future native-glove support.

Use the web interface

Page	What it does
Dashboard, `/dashboard`	Shows the camera and generated inputs; selects the current profile and starts or stops delivery.
Glove Academy, `/learn`	Provides twelve practice lessons and guided gesture tuning, with game input paused.
Help, `/help`	Opens the local manuals and PDFs; **This cabinet** shows current connection details.
Setup, `/setup`	Saves connection, camera, and startup settings; the Games section edits RetroPie mappings with backup and restore. Pairing requires HTTPS on port 8443.

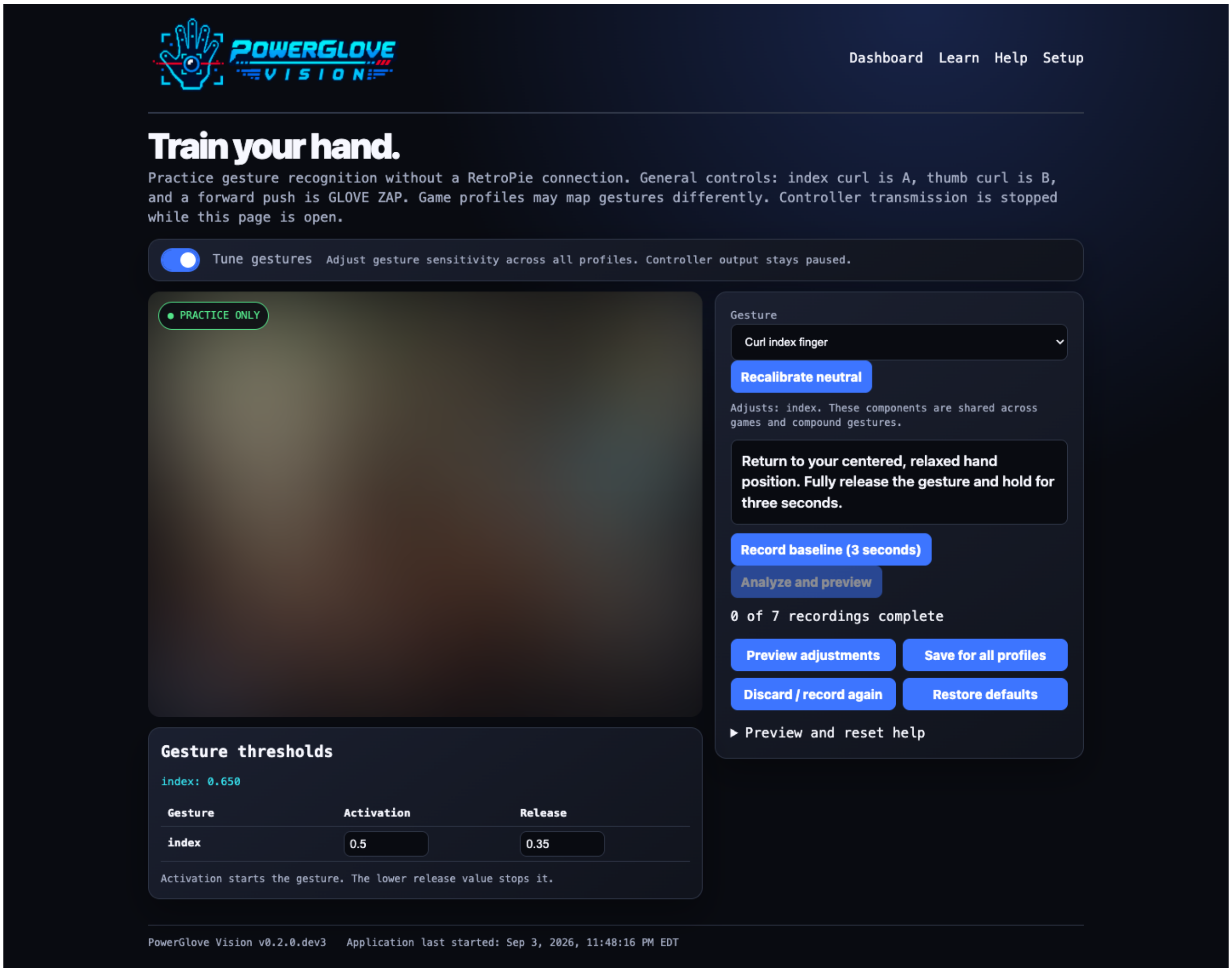
With **Gestures off** selected, the camera stays closed. Choose an active profile or open Glove Academy to begin. Wait for the camera view before practicing or playing; starting immediately after a reboot can take longer.

Stop controller pauses delivery while leaving active tracking available. **Gestures off** closes the camera. **Shutdown** requests a Linux halt, but the tested UNO Q restarts afterward. A disappearing website is not proof that it is safe to remove power. See the installation guide before using Shutdown.



Dashboard showing the selected profile and controller readings

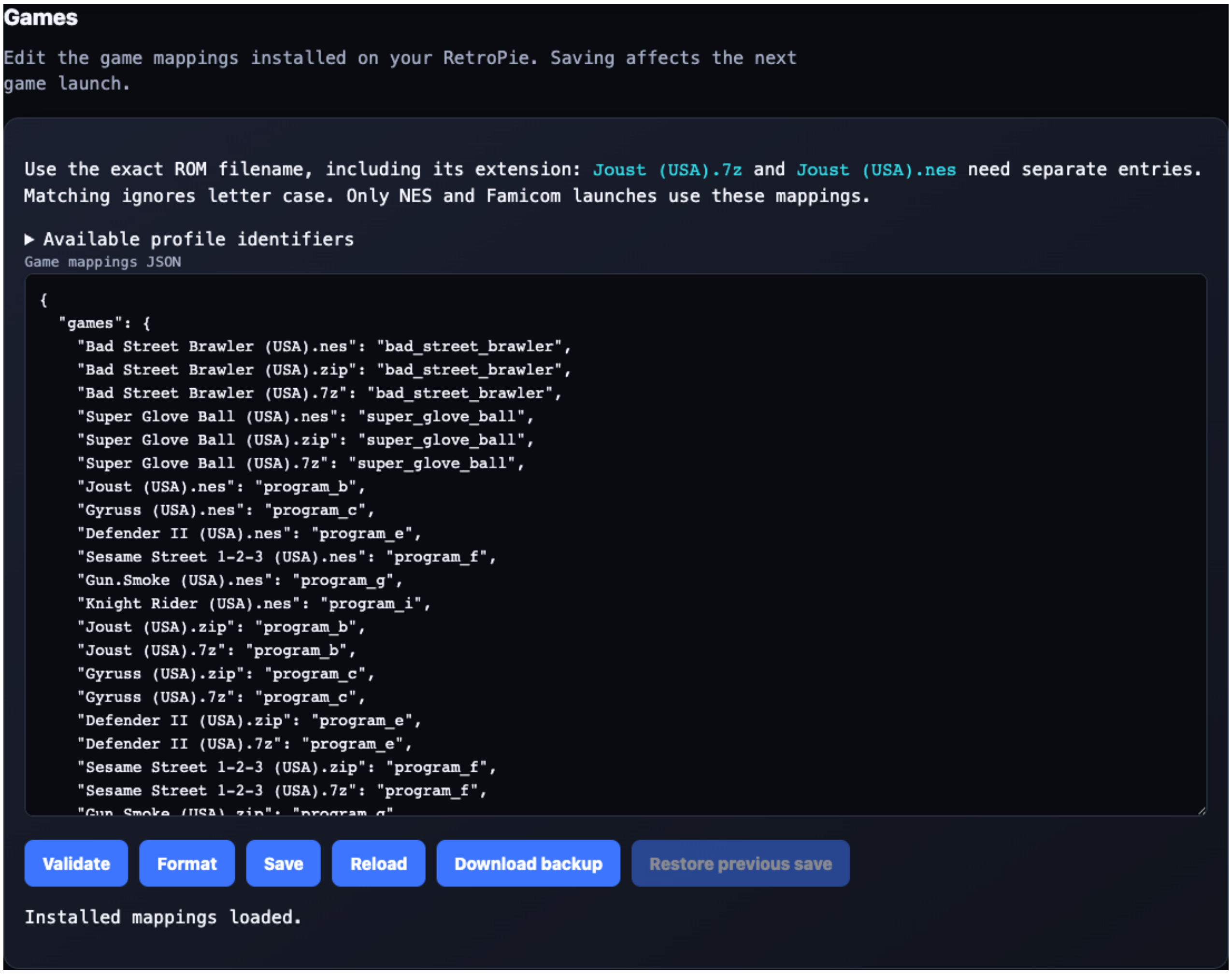
The screenshots below show the current interface. Camera imagery is blurred for privacy.



Glove Academy in Tune mode, with thresholds below the blurred camera

In **Glove Academy**, switch on **Tune gestures** to adjust sensitivity. The UNO Q shows a scanning **T** during tuning and a matching scanning **L** during ordinary practice. Both modes pause game input.

Tuning uses three recordings of three seconds each: open hand, gesture, open hand. Optional **Set up my hand** uses open hand, gentle fist with the thumb outside, open hand to measure all five fingers. Preview before saving. Saved Activation and Release thresholds apply in gameplay across profiles. **Glove Zap** and **Pull Back** have separate practice lessons and tuning controls; movement tuning ends by returning to the starting position and camera distance.



Games editor in the lower part of Setup

Scroll down **Setup** to **Games** to map exact ROM filenames to profiles. Saving affects the next game launch, not the game already running.

Maintain or extend the project

Use the [Installation Guide's maintenance section]() for updates, and the [Configuration Reference]() for settings and all command options. The [Contributing guide]() covers tests, documentation, package verification, and releases. Printable editions are stored in [output/pdf/]().

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